Air Lift Column Material Selection

Criterion	HDPE – High density polyethylene	PVC (Polyvinyl Chloride)	Notes/ Consideration (Pros and Cons)
Cost	Moderate in cost for small size	Low (often the cheapest option)	- HDPE - A Better option for larger diameters - PVC - Very cost effective for typical pipe sizes
Chemical Resistance	Can withstand strong UV light. Excellent resistant to strong acids and bases	Good (affected with very strong UV light. High resistance to phosphates and nitrates ions.	- Both are excellent in resistivity to phosphorus and nitrogen levels in wastewater.
Structural Strength	Good (flexible and strong but can sag if not supported properly).	Very good (rigid and holds shape, but brittle in cold temperatures)	- HDPE requires more support for long horizontal runs - PVC is excellent for rigid vertical columns
Ease of Fabrication	Moderate (requires heat welding) Considered more complex and requires training.	Easy (uses common solvent welding/gluing). Can be done without training	- HDPE: Stronger welds once done, but more specialized tools. - PVC: Simple gluing, DIY friendly.
Durability	Excellent (long lifespan and good UV resistance)	Good (long lifespan, can become brittle with frequent UV exposure)	- HDPE: Better for outdoor/sun exposure

			without special treatment. - PVC: Might need paint/UV protection if exposed.
Sustainability	Good recycling capability, sometimes readily available with recycled content	Moderate ability to recycle, with concerns of chlorine content in manufacturing and disposal.	- Consider local recycling infrastructure for both.
Non-toxicity to algae	Excellent (inert and stable polymer and widely used in food grade applications due to its low leaching of harmful chemicals)	Good (safe for water systems, rigid form has minimal leaching concerns, widely used).	- both HDPE and rigid PVC are considered safe and non-toxic for algae growth.

Justification

- PVC was chosen as the best candidate due to its significantly low cost and widespread availability.
- It is considered an economically viable choice for large-scale systems.
- Has excellent rigidity and structural strength to ensure stability for tall structural columns
- It's ease of fabrication through solvent welding allows straightforward construction without much complication
- Furthermore, PVC offers good chemical resistance to wastewater chemicals.
- Is non-toxic to algae, ensuring system integrity and biological compatibilty